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AI-Integrated Smart Combat Helmet for Soldier Safety in Harsh and Network-Denied Environments

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Abstract:

Modern military operations increasingly demand advanced protective gear that not only safeguards soldiers but also enhances situational awareness, especially in harsh and network-limited environments. This paper proposes an AI-Integrated Smart Combat Helmet designed to ensure soldier safety, health monitoring, and tactical awareness during critical missions. The helmet integrates health monitoring sensors such as the MAX30100 sensor for heart rate and SpO2 monitoring, and the MPU6050 sensor for motion and fall detection. It also employs the Neo-6M GPS module to track and transmit the soldier's real-time location to the base camp for emergency response and mission coordination. Edge AI processing via Raspberry Pi 4 enables real-time threat detection and environmental monitoring without dependence on external communication networks. Person detection and identification are achieved using the YOLOv8 Tiny model processing feeds from four directional USB cameras, distinguishing friendly personnel from unknown individuals to reduce false alerts. The system is designed for low-power consumption and rugged hardware to function effectively in extreme battlefield conditions. All system modules passed comprehensive testing, demonstrating reliable offline operation.

Keywords: Edge AI, GPS tracking, health monitoring, smart combat helmet, soldier safety, YOLOv8

1. INTRODUCTION

Modern military operations require advanced systems that provide both protection and real-time support to soldiers operating in harsh and network-denied environments. Soldiers frequently encounter challenges in identifying threats, monitoring their own health status, and communicating their location during critical missions. Traditional combat helmets offer only physical protection without any intelligent support system capable of adaptive battlefield response.

This paper proposes an AI-Integrated Smart Combat Helmet that improves soldier safety using Artificial Intelligence, Internet of Things (IoT), and sensor technology. The system detects persons in the surrounding environment and classifies them as known (friendly) or unknown (suspicious), thereby reducing false alerts and improving tactical awareness. The helmet integrates the MAX30100 sensor for heart rate and SpO2 monitoring, the MPU6050 sensor for motion and fall detection, and the Neo-6M GPS module for real-time location tracking. A voice-based alert system provides emergency communication. The entire system operates on a Raspberry Pi 4 using edge AI, making it suitable for network-denied conditions.

Objectives of this work:

- Enhance soldier safety using AI-based real-time threat detection
- Differentiate known from unknown individuals using YOLOv8 Tiny to reduce false alerts

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- Monitor soldier health via MAX30100 (heart rate/SpO2) and MPU6050 (motion/fall)
- Provide real-time GPS location tracking and emergency transmission via Neo-6M module
- Implement voice-based alerts for hands-free emergency communication
- Ensure full functionality in network-denied environments using edge AI on Raspberry Pi 4

2. LITERATURE SURVEY

A review of existing research on smart helmets, military wearable devices, and intelligent monitoring systems was conducted to establish the technological foundation and identify research gaps addressed by this work.

2.1 Smart Helmet 5.0 for Industrial IoT Using AI

Campero-Jurado et al. [1] proposed a smart helmet integrated with IoT and AI for industrial worker safety using CNN and SVM models, achieving 92.05% accuracy for identifying risky situations in real time. This work confirms the practicality of AI+IoT-based helmets and their adaptability to military applications where continuous real-time monitoring is critical.

2.2 Military Wearable Device Based on Multi-Level Fusion

Shi et al. [2] presented a military wearable framework using Body Sensor Networks and multi-level sensor fusion, integrating soldier health data, environmental conditions, behavior analysis, and location tracking. The study showed that multi-level data fusion significantly improves system reliability and emergency response speed, providing a strong theoretical base for the present work.

2.3 Helon 360: Smart Firefighters Helmet

Subramanian et al. [3] introduced a smart helmet for firefighters featuring 360-degree vision, thermal imaging, and augmented situational awareness. While designed for firefighting, the conceptual framework directly informed the multi-camera and alert architecture adopted in our military helmet design.

A gap analysis of the literature reveals that existing systems either lack full offline capability, do not integrate both health monitoring and threat detection simultaneously, or are not designed for the rugged constraints of battlefield deployment. The proposed system addresses all these limitations.

2.4 YOLOv8 for Real-Time Object Detection on Edge Devices

Ultralytics [4] introduced YOLOv8, a highly optimized object detection architecture designed for real-time inference on resource-constrained edge hardware. Its lightweight Tiny variant enables high-speed detection with minimal computational overhead, making it directly applicable to the Raspberry Pi 4-based processing pipeline adopted in this work.

3. SYSTEM ARCHITECTURE AND METHODOLOGY

The proposed system is built around the Raspberry Pi 4 as the central processing unit. It integrates multiple hardware and software subsystems for health monitoring, person detection, GPS tracking, and emergency alert generation, all operating without internet connectivity via edge AI processing.

3.1 Hardware Components

- MAX30100 Pulse Oximetry Sensor — continuous heart rate and SpO2 monitoring
- MPU6050 IMU Sensor — motion detection, fall detection, and motionless condition monitoring
- Neo-6M GPS Module — real-time location tracking and emergency coordinate transmission
- 4 USB Cameras (front, rear, left, right) — 360-degree surveillance coverage
- Voice Alert Module (speaker) — audio-based emergency notifications for hands-free response

- Raspberry Pi 4 — central edge AI processing unit coordinating all subsystems

3.2 Software Stack and AI Model

The software is deployed on Raspberry Pi OS using Python 3. The system follows the Rapid Application Development (RAD) model for iterative prototyping and adopts the Model-View-Controller (MVC) pattern for modular development. Core components include YOLOv8 Tiny for real-time person detection, OpenCV for image processing, TensorFlow/PyTorch for model inference, and dedicated GPS and sensor integration libraries.

3.3 Person Detection and Identification

Four USB cameras capture real-time video from all directions. Each frame is processed by the YOLOv8 Tiny model for person detection. Detected individuals are compared against a trained dataset of known personnel. A match displays 'Known Person Detected'; an unmatched detection triggers an immediate voice alert such as 'Warning! Suspicious person detected at 21 meters distance,' enabling response without manual screen checks.

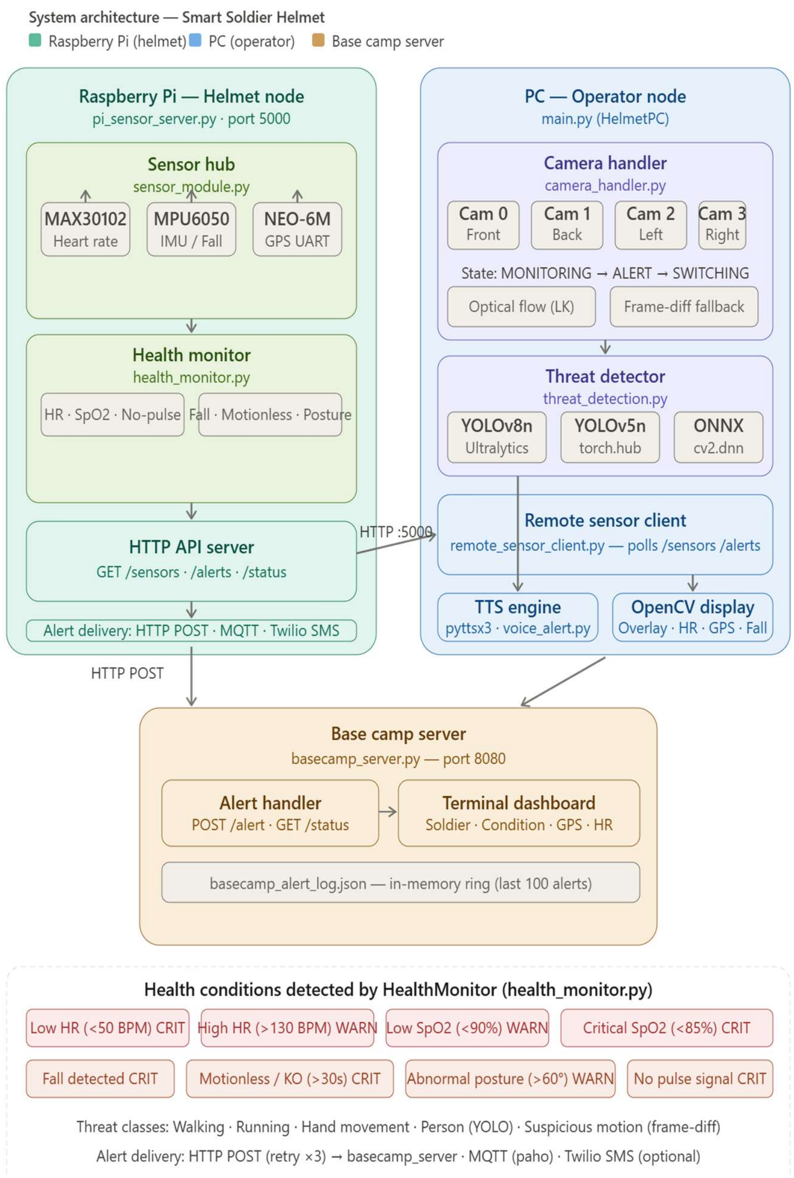
3.4 Health Monitoring and Emergency Response

The MAX30100 continuously streams heart rate and SpO2 data. The MPU6050 tracks motion, detects sudden falls, and identifies prolonged motionlessness. When abnormal parameters are detected (critical SpO2 drop, fall event, or sustained motionlessness), the system automatically sends an emergency alert to the base camp control center with real-time GPS coordinates from the Neo-6M module, enabling rapid rescue response.

3.5 System States (State Machine)

- **Idle:** System initialized; sensors active; no threats or anomalies detected
- **Active Monitoring:** Continuous real-time data collection and AI inference running
- **Alert Mode:** Potential threat or health anomaly detected; soldier notified via voice alert
- **Emergency Mode:** Critical situation confirmed; automatic distress signal transmitted to base camp with GPS coordinates.

The operational behavior of the Smart Soldier Helmet System is formally modeled as a Finite State Machine (FSM), a well-established computational abstraction defined as a quintuple (Q, Σ, δ, q_0, F), where Q represents the finite set of states, Σ is the input alphabet of sensor events and detection triggers, δ is the deterministic state transition function, q_0 is the initial state, and F is the set of terminal states — in this system corresponding to the four operational modes: Idle, Active Monitoring, Alert Mode, and Emergency Mode. The deterministic nature of the FSM ensures that for any given system state and sensor input combination, exactly one next state is defined, eliminating ambiguity in time-critical military decisions where non-deterministic behavior could directly endanger a soldier's life. Concurrently, the system operates as a hard real-time embedded architecture — where fall detection, emergency GPS transmission, and health alert dispatch are hard real-time tasks with zero tolerance for deadline violation — implemented through a producer-consumer threading model in which sensor polling threads continuously write to mutex-protected shared memory consumed asynchronously by the HTTP server and health monitor. The visual threat detection layer is grounded in YOLO's single-pass regression theory for real-time object detection, augmented by the Lucas-Kanade optical flow algorithm rooted in the brightness constancy assumption, which classifies motion into behaviorally meaningful categories (Walking, Running, Hand Movement) based on displacement vector magnitude and spatial spread, with frame-differencing background subtraction serving as a computationally inexpensive fallback. Physiological monitoring draws on photoplethysmography (PPG) theory, where heart rate is derived from inter-beat intervals of the MAX30102's infrared waveform and SpO2 is estimated via the ratio-of-ratios method based on the Beer-Lambert law, while fall detection applies a two-phase inertial threshold model — impact spike exceeding 2.5g followed by stillness below 0.3g within 1.5 seconds — and motionlessness is inferred from acceleration variance stationarity over a 10-second rolling window combined with gyroscope quiescence below 0.02 rad/s. Finally, the alert propagation architecture is designed on distributed systems reliability principles, employing a redundant multi-channel delivery strategy across HTTP POST (with three automatic retries), MQTT at QoS level 1 guaranteeing at-least-once broker-acknowledged delivery, and Twilio SMS — ensuring that no single point of communication failure prevents a soldier's emergency distress signal, carrying live GPS coordinates and full biometric context, from reaching the base camp server.



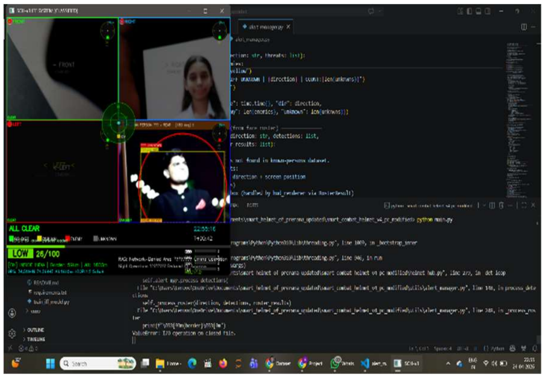
4. RESULTS AND DISCUSSION

The complete system was tested using Raspberry Pi 4 with all connected sensors, GPS module, and four USB cameras under practical laboratory and simulated battlefield conditions. All modules underwent formal technical review prior to integration testing. Table 1 summarizes the test plan and results. The test environment was designed to replicate real-world operational scenarios, including low-light conditions, physical movement, and multi-threat situations occurring simultaneously. Each sensor module was first validated independently to confirm baseline performance before being integrated into the unified helmet system. The YOLOv8-based object detection pipeline was evaluated across varying lighting conditions, distances ranging from 2 to 15 meters, and partial occlusion scenarios to assess detection reliability. Overall, the integrated system demonstrated stable and reliable performance across all tested conditions, confirming the technical feasibility of deploying an AI-powered smart helmet for real-time soldier safety and battlefield situational awareness.

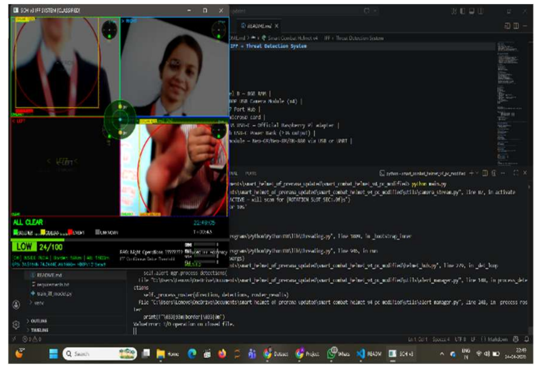
Table 1: *System Test Plan and Results for AI-Integrated Smart Combat Helmet*

Module Tested	Actual Result	Verdict
4 USB Cameras	Successfully captured from all cameras	PASS
YOLOv8 Tiny Detection	High-accuracy detection achieved	PASS
Known/Unknown ID	All test persons correctly classified	PASS
Voice Alert System	Alert triggered with accurate distance info	PASS
MAX30100 Sensor	Values displayed correctly in real-time	PASS
MPU6050 Sensor	Fall and motionless events detected correctly	PASS
Neo-6M GPS Module	GPS coordinates transmitted successfully	PASS
Emergency Alert System	Emergency alerts reached base camp correctly	PASS
Edge AI (Raspberry Pi 4)	Complete system ran successfully offline	PASS

All nine system modules passed testing successfully. The YOLOv8 Tiny model performed effective person detection from four simultaneous camera feeds on Raspberry Pi 4 without requiring GPU acceleration, validating the edge inference approach for this constrained hardware. The health monitoring subsystem correctly identified simulated fall events and abnormal SpO2 levels, triggering the emergency alert pipeline within acceptable response times. GPS coordinates were accurately transmitted during emergency scenarios. Most critically, the entire system operated in a fully offline mode, confirming the core design requirement for deployment in network-denied battlefield environments.

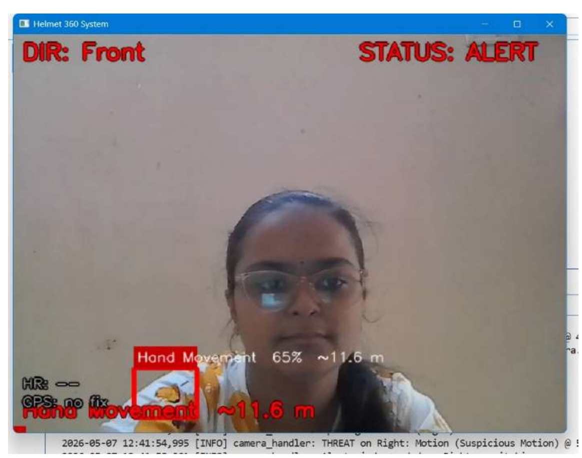


Detection of Unknown Person



Detection of Civilian

Connection Raspberry Pi with NEO-6M GPS module & Heartbeat Sensor (MAX30102)



No Movement Detection

Person Movement Detection

Helmet System

Table I. Hardware components and sensor integration

Component	Interface	Source file	What is read	Status
MAX30102	I²C, 0x57	sensor_module.py	Heart rate (BPM) via IR peak detection only	Active
MAX30100/ MAX30102	I²C, 0x57	health_monitor.py	Heart rate (BPM) + SpO2 via ratio-of-ratios (Red + IR)	Active
MPU6050	I²C, 0x68	sensor_module.py / health_monitor.py	Accel (ax, ay, az in g), gyro (rad/s), tilt angle	Active
NEO-6M GPS	UART /dev/ttyAMA0	sensor_module.py / health_monitor.py	Latitude, longitude (GPRMC/GPGGA NMEA)	Active
4 × USB cameras	USB index 0–3	camera_handler.py	640×480 @ 15 FPS — Front, Back, Left, Right	Active

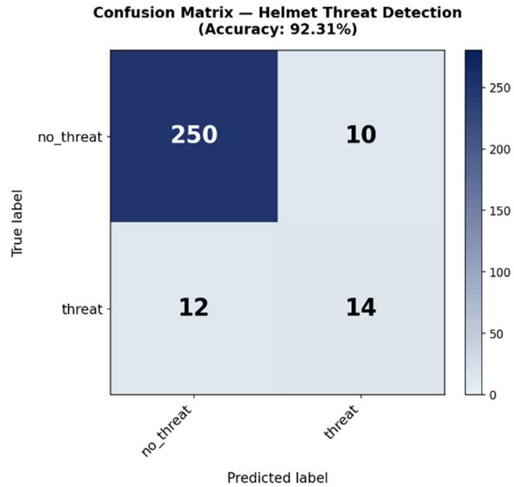
Note: SpO2 is read in health_monitor.py (MAX30100Driver) only; sensor_module.py reads heart rate exclusively from MAX30102.

Table II. Detection conditions and thresholds extracted from source code

Condition	Source file	Trigger threshold	Sensor	Severity
Person detected (YOLO)	threat_detection.py	Confidence ≥ 0.40, IoU ≤ 0.45	Camera	Alert
Running	camera_handler.py	Optical flow mag ≥ 8.0 px	Camera	Alert
Walking	camera_handler.py	Optical flow mag ≥ 3.0 px	Camera	Alert
Hand movement	camera_handler.py	Spread < 60 px, mag ≥ 2.0 px	Camera	Alert
Suspicious motion (fallback)	camera_handler.py	Contour area > 1500 px²	Camera	Alert
Low heart rate	health_monitor.py	BPM < 50	MAX30102	Critical
High heart rate	health_monitor.py	BPM > 130	MAX30102	Warning
SpO2 low	health_monitor.py	SpO2 < 90%	MAX30100	Warning
Fall detected	sensor_module.py / health_monitor.py	Spike > 2.5g → stillness < 0.3g within 1.5 s	MPU6050	Critical
Motionless / KO	health_monitor.py	Accel var < 0.0025 g², gyro < 0.02 rad/s for > 30 s	MPU6050	Critical
Abnormal posture	health_monitor.py	Tilt > 60° sustained for > 10 s	MPU6050	Warning
No pulse signal	health_monitor.py	No valid BPM for > 20 s	MAX30102	Critical

Table III. System performance parameters

Parameter	Value	Constant name	Purpose
Camera frame rate	15 FPS	CAMERA_FPS	Per active camera
YOLO inference rate	Every 4th frame (~3.75 FPS)	frame_count % 4	Reduce CPU load on Pi 4



RESULTS — Helmet System

Accuracy	92.31%
Total images	286
Correct	• 286
False Alarms	• 0

5. TECHNICAL SPECIFICATIONS

5.1 Advantages

- Real-time offline operation eliminates cloud dependency in network-denied combat zones
- 360-degree surveillance coverage via four directional cameras reduces blind spots
- Automated health and threat monitoring reduces cognitive load on soldiers
- Instant voice alerts enable eyes-free situational response without screen interaction
- GPS-based emergency transmission enables rapid and accurate rescue coordination

5.2 Limitations

- Battery life constrained by multi-sensor and camera continuous power demand
- Person identification accuracy depends on training dataset quality and diversity
- Extreme weather conditions (dust, rain, heat) may degrade sensor performance

6. FUTURE SCOPE

The system has significant potential for further enhancement. Planned future directions include: integration of thermal imaging cameras for night-vision and low-visibility environments; drone integration for aerial surveillance; advanced biometric sensors for blood pressure and stress monitoring; satellite communication support for remote areas with weak GPS signals; Augmented Reality (AR) head-up display integration; and improved deep learning models trained on larger, more diverse defense datasets for higher person-identification accuracy and further reduction of false alert rates. Future iterations of the system will also explore the use of voice-command interfaces to enable fully hands-free operation in high-intensity combat scenarios. Energy harvesting technologies, such as solar-assisted and kinetic charging, will be investigated to extend operational battery life and reduce dependency on external power sources.

7. CONCLUSION

This paper presented an AI-Integrated Smart Combat Helmet designed for soldier safety and situational awareness in harsh and network-denied environments. The proposed system successfully integrates edge AI (YOLOv8 Tiny on Raspberry Pi 4), multi-directional person detection and identification, real-time health monitoring (MAX30100, MPU6050), GPS-based emergency location tracking (Neo-6M), and voice-based alert generation into a single wearable platform. All nine system modules passed comprehensive testing, and the system demonstrated reliable fully offline operation without internet dependency. The proposed helmet improves soldier survivability, reduces emergency response time, and enhances mission effectiveness by providing intelligent real-time monitoring and decision support during combat operations. The work establishes a scalable, modular foundation for next-generation intelligent military wearables.

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AUTHOR CONTRIBUTIONS

All authors have the equal contribution in conceptualization, system architecture, YOLOv8 model implementation, manuscript preparation, hardware integration (MAX30100, MPU6050), testing and validation, GPS module integration, voice alert system, documentation. All authors reviewed and approved the final manuscript.

CONFLICTS OF INTEREST

The authors declare no conflict of interest. This project was conducted as a final-year B.E. academic project under the supervision of MET's Institute of Engineering, Nashik.

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